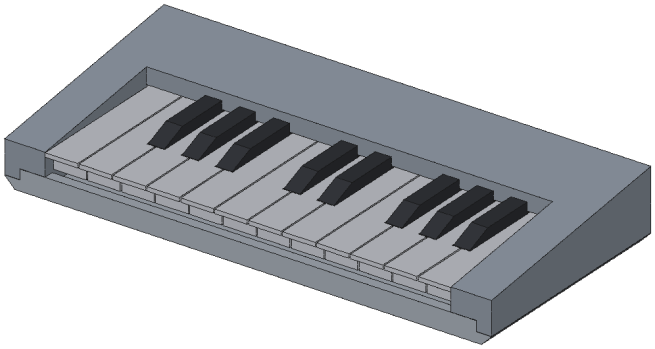
Operations Sequence: Metal MIDI Keyboard

ME 103 · Alex Hong · October 21, 2026

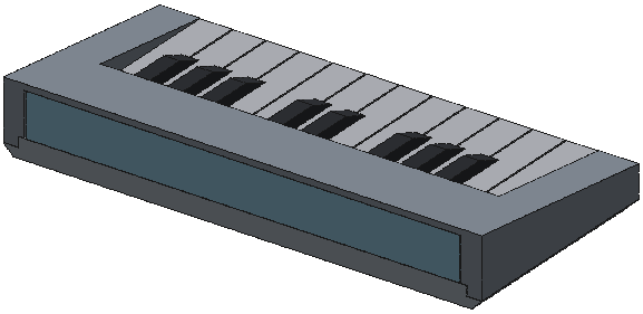
A 20-key (E to B: 12 white, 8 black) MIDI keyboard made entirely of 356A aluminum. The frame (Top Part), Base and key blanks are sand cast in the PRL foundry from 3D-printed patterns, then finished on the manual mill. The keys pivot on a purchased Ø.250 stainless rod, and a glass window sits in the back wall. In the images, **red** is the material removed in that step and **dark gray** is the workholding.

#	To be fixed in CAD before making patterns
1	**Glass slot → cast rabbet.** The enclosed .100 x 1.10 slot can't be sand cast and needs a slitting saw. Replace it with a .110-deep step on the outer back face (X .525–11.475, open at the bottom, up to Z 1.35), delete the Base glass slot, and resize the glass to 1.10 x 10.85 x .098. This sequence already assumes the rabbet.
2	**Top skin too thin.** The sloped skin over the keys is .044–.056 thick; thicken to ~3/16 so the casting fills.
3	**Pivot holes are both blind.** The 11.00 rod can't be installed. Make the right-cheek hole go through.
4	**No end clearance on the key stack.** Keys + spacers fill the 10.000 cheek gap exactly; leave ~.010 so keys don't bind.
5	**Short screw engagement.** 4-40 x 1/4 reaches only ~.10 into the Top Part; 5/16 long (92196A107) is better.

CAD

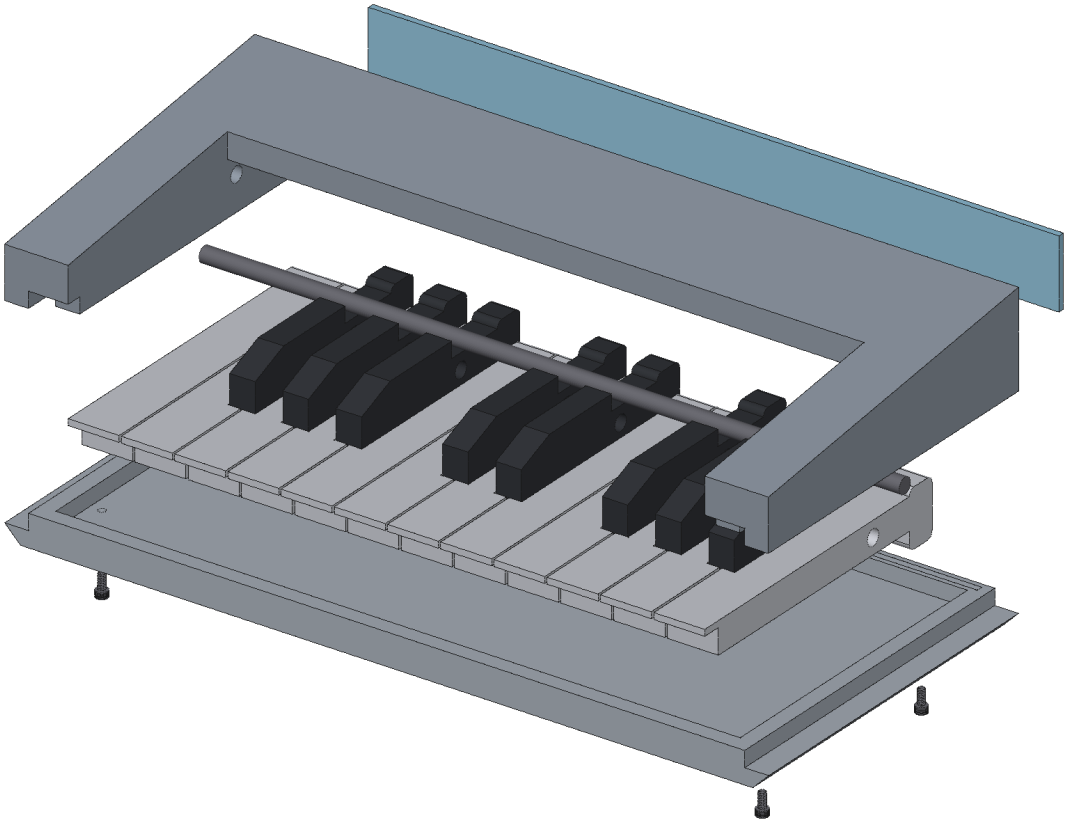


Assembly, front

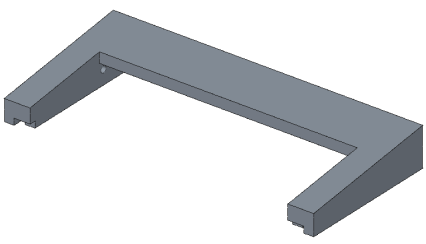


Assembly, back (glass window)

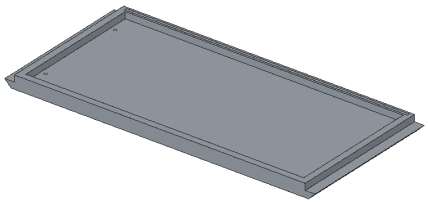
Exploded assembly



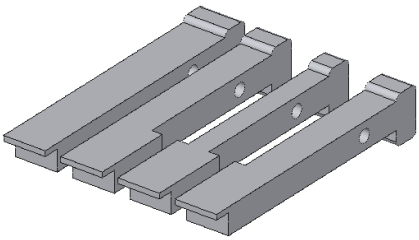
Item	Qty	Part	Make / buy
1	1	Top Part (frame)	356A, cast + milled
2	1	Glass window 1.10 x 10.85 x 2.5 mm	Buy, cut to size
3	1	Pivot rod Ø.250 x 11.00, 303 SS	McMaster 1257K66, cut
4	12	White keys	356A, cast bar + milled
5	8	Black keys	356A, cast bar + milled
6	1	Base	356A, cast + milled
7	4	4-40 x 1/4 SHCS, 18-8 SS	McMaster 92200A106
8	3 / 16	Spacers .036 / .018, 1/4 ID	McMaster ring shims
9	20	Felt pads	Master Caster 88496



Top Part

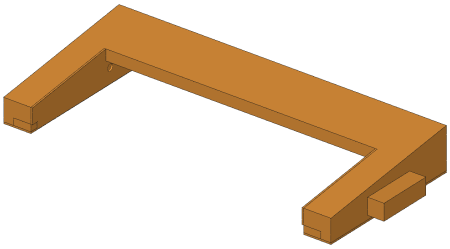


Base



White keys: plain, right notch, both, left notch

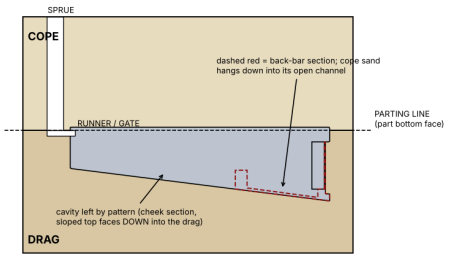
Top Part (frame): sand cast 356A, then milled



1. Print pattern

PLA pattern = part + .060 machining stock, 2° draft, x1.013 shrink. Glass rabbet and window cast in; pivot holes filled. Riser on the right cheek.

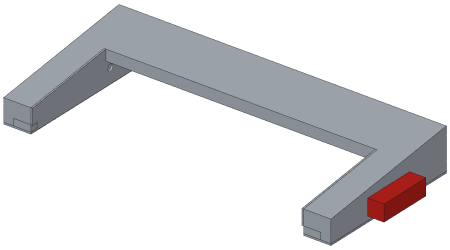
Tooling: 3D printer, PLA (tooling only)



2. Mold and pour

Pattern bottom-face at the parting line, whole part in the drag. Ram, draw pattern, gate, pour ~2.4 lb 356A at ~1,350 °F.

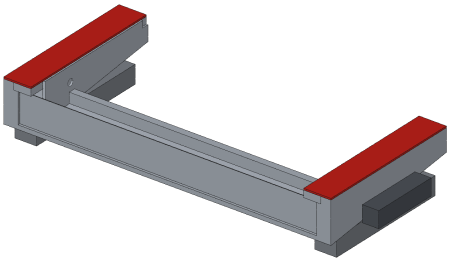
Machine: PRL foundry
Tooling: Oil-bonded sand, cope + drag flask



3. OP 10 Cut off riser

Cut riser and gate off, sand flush. Inspect for porosity.

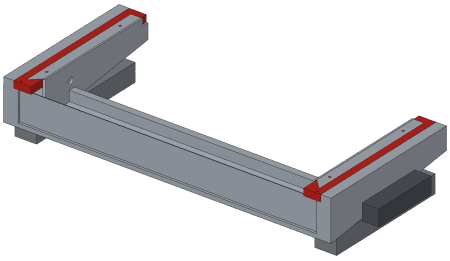
Machine: Bandsaw
Workholding: Bandsaw vise
Tooling: Bi-metal blade, 10-14 TPI



4. OP 20 Face bottom (Datum A)

Face both cheek bottoms .060; all later setups locate off this face.

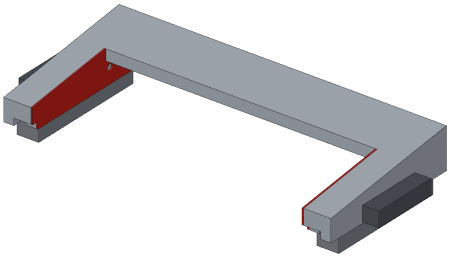
Machine: Manual mill
Workholding: Upside down on 7.1° wedge parallels, toe clamps
Tooling: 2.5" fly cutter



5. OP 30 Relief grooves and locating blocks

Same setup. Mill the .25 x .25 grooves and the .375-wide blocks that nest in the Base pocket.

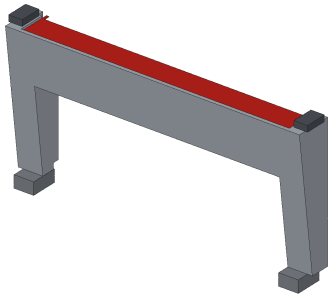
Machine: Manual mill
Workholding: Same as OP 20
Tooling: 1/4" end mill, edge finder



6. OP 40 Inner cheek faces

Side-mill both inner faces to a 10.000 gap, square to Datum A.

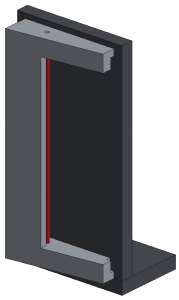
Machine: Manual mill
Workholding: Datum A on parallels, toe clamps
Tooling: 1/2" end mill



7. OP 50 Clean up glass rabbet

Skim the cast rabbet seat .020 to a flat .110 depth; square the ends 10.95 apart.

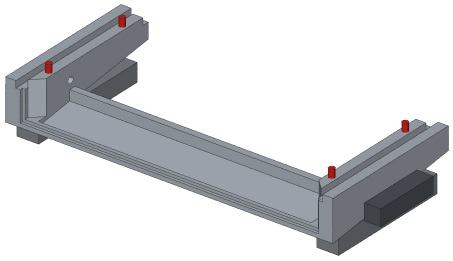
Machine: Manual mill
Workholding: On front face, back face up, strap clamps
Tooling: 1/2" and 1/4" end mills



8. OP 60 Pivot holes

Ø.257 through the right cheek, across the gap, .50 into the left cheek in one pass so the holes are coaxial.

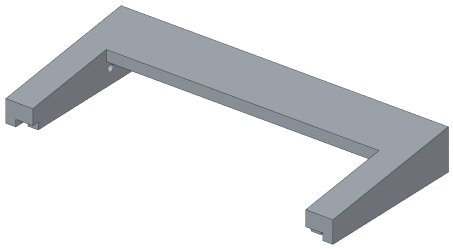
Machine: Manual mill
Workholding: Stood on end on an angle plate
Tooling: Center drill, letter-F extension drill



9. OP 70 Tapped holes

4x #43 drill .40 deep, tap 4-40 (transfer from the finished Base).

Machine: Manual mill
Workholding: Same as OP 20
Tooling: #43 drill, 4-40 tap

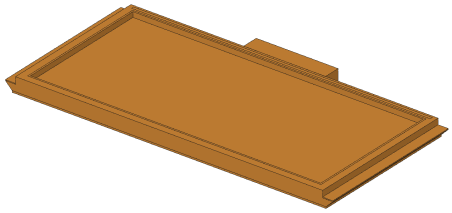


10. OP 80 Deburr and finish

Deburr, sand the top 220 → 400, bead blast. Check rod and glass fit.

Machine: Bench
Tooling: Files, deburr tool, bead blaster

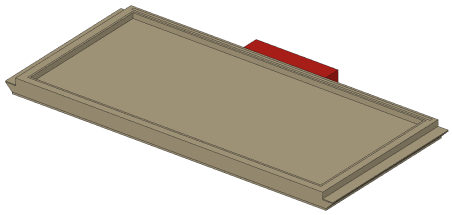
Base: sand cast 356A, then milled



1. Print pattern

PLA pattern = part + .060 stock on bottom and pocket, 2° draft. Pocket faces down in the drag; end shelves and chamfers cast in.

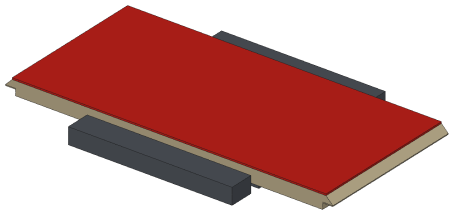
Tooling: 3D printer, PLA (tooling only)



2. OP 10 Pour and cut off riser

Pour ~4 lb 356A; cut off riser and gate. Check the .25 floor for warping.

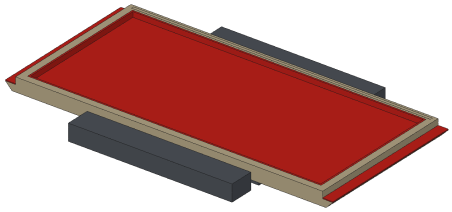
Machine: Foundry, bandsaw
Tooling: Bi-metal blade



3. OP 20 Face bottom (Datum A)

Pocket side down; face the bottom .060 flat.

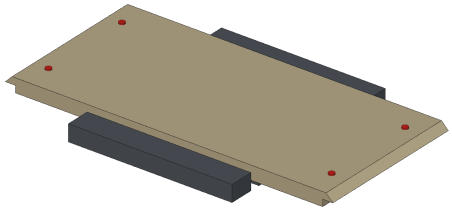
Machine: Manual mill
Workholding: 6" vise on parallels, ends supported
Tooling: 2.5" fly cutter



4. OP 30 Pocket and shelves

Finish pocket to 10.75 x 5.00 x .25 deep (+.005/side) and skim end shelves.

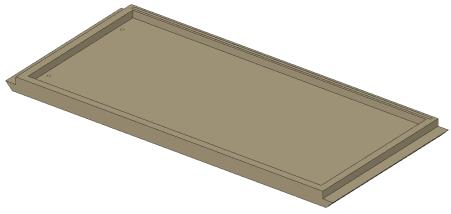
Machine: Manual mill
Workholding: Datum A down in vise
Tooling: 1/2" and 1/4" end mills



5. OP 40 Screw holes

4x Ø.116 through, counterbore Ø.218 x .12 for flush 4-40 heads.

Machine: Manual mill
Workholding: Flipped in vise
Tooling: #32 drill, 7/32" counterbore



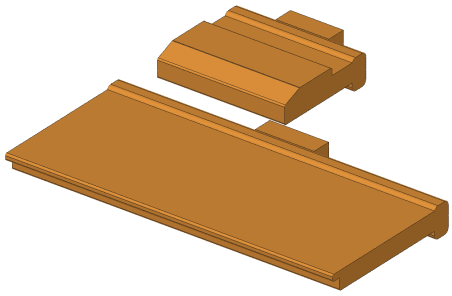
6. OP 50 Deburr and finish

Deburr, file pocket corners, bead blast.

Machine: Bench

Keys: cast profile bars, milled together, then cut apart

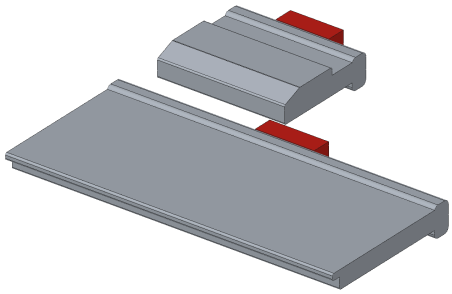
All 12 white keys are machined as one 11.0" bar and all 8 black keys as one 4.2" bar, so every key has the same profile. The bars are then sawed into keys.



1. Print patterns

Split PLA patterns of the white and black profile bars (+.060 stock, 2° draft), split at the plate bottom.

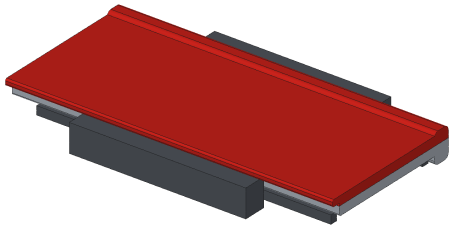
Tooling: 3D printer, PLA (tooling only)



2. OP 10 Pour and cut off risers

Pour both bars on one gating tree (~7 lb 356A); cut off risers.

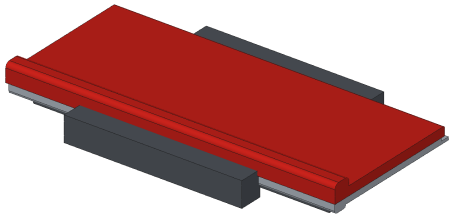
Machine: Foundry, bandsaw
Tooling: Bi-metal blade



3. OP 20 Mill top of profile

Face the playing surface to .800 and counterweight top to .900; square bar ends; round counterweight edges.

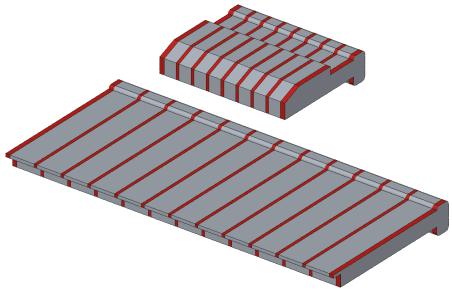
Machine: Manual mill
Workholding: Vise on parallels, jaws on back face and under front lip
Tooling: 1/2" end mill, 3/32" corner rounder



4. OP 30 Mill underside

Flip. Mill plate bottom (key .500 thick), counterweight bottom, front-lip notch, and front/back faces.

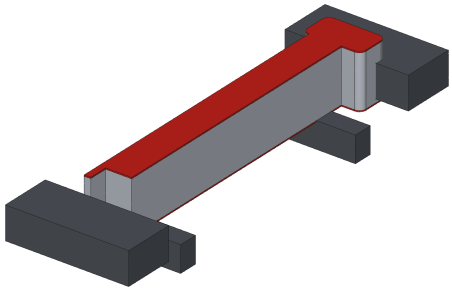
Machine: Manual mill
Workholding: Same vise setup, flipped
Tooling: 1/2" and 1/4" end mills



5. OP 40 Saw into keys

Cut at .900 pitch (white) and .500 pitch (black), leaving ~.020 per side.

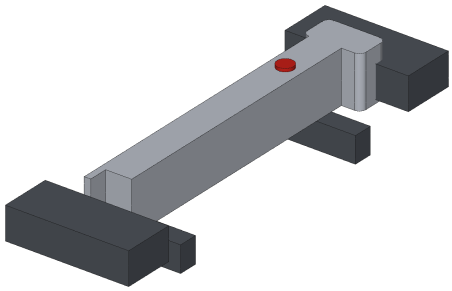
Machine: Bandsaw
Workholding: Bandsaw vise with stop block
Tooling: Bi-metal blade



6. OP 50 Face to width

Face both sides of each key to .800 (white) / .400 (black).

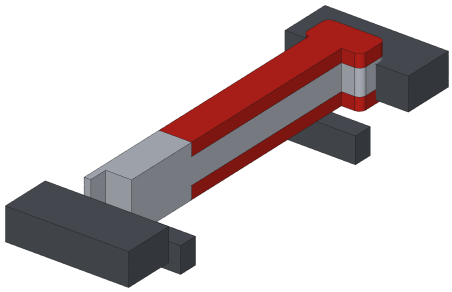
Machine: Manual mill
Workholding: Key on its side in vise, vise stop
Tooling: Fly cutter



7. OP 60 Drill pivot holes

Ø.257 at Y 3.700, Z .550 in all 20 keys, same setup, so the holes line up.

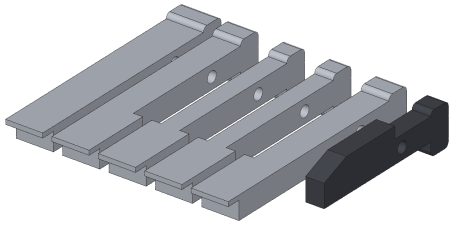
Machine: Manual mill
Workholding: Same as OP 50
Tooling: Center drill, letter-F drill



8. OP 70 Black-key notches

Mill .200 steps from Y 1.500 back: right side on F, C, F; left on B, E, B; both on G, A, D, G, A.

Machine: Manual mill
Workholding: Same as OP 50
Tooling: 1/2" end mill

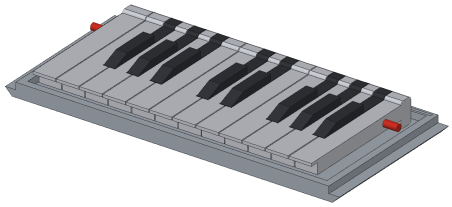


9. OP 80 Deburr and finish

Deburr, sand playing surfaces, bead blast; optionally darken the black keys.

Machine: Bench

Assembly



1. Assemble

(1) Set the glass into the rabbet with clear RTV. (2) Stack keys and spacers on the rod: .036 between E-F, B-C, E-F; .018 on both sides of each black key. (3) Push the rod through the right cheek into the left blind hole; retaining compound on the end. (4) Set the frame on the Base and fasten 4x 4-40 screws from below. (5) Stick felt pads under the counterweights.